Welcome to MULT20015

“Elements of Quantum Computing”

University of Melbourne

Staff:

Physics: Dr. Josh Combes, Prof. Lloyd Hollenberg, A/Prof. Muhammad Usman, Dr. Lucas Hackl

Computer Science: Prof. Adrian Pearce

Finance: Dr. Nitin Yadav

MULT20015 Elements of Quantum Computing

This subject is a relatively gentle introduction to quantum computing and applications for students from a range of backgrounds.

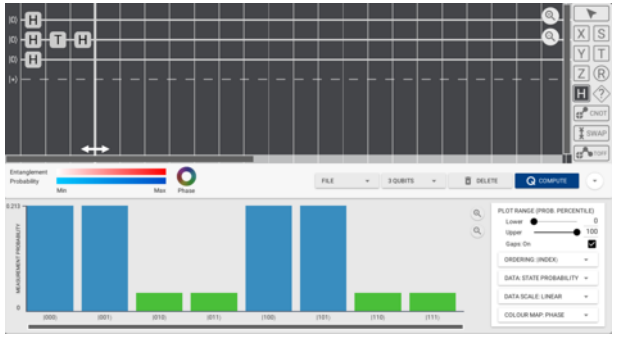
In this subject we will cover:

- The basic concepts of quantum information and quantum logic
- Programming quantum computers to perform computational tasks
- How quantum computers can be applied to problems in the context of mathematics, computing and finance
- The status of current quantum computer technology

The aim is to give you an overview of quantum computing (and some related topics)
-> position for further studies in quantum computing
(e.g. Masters: MULT90063 *Intro to QC,* COMP90084 *Quantum Software Fundamentals*)

Teaching system: UoM Quantum User Interface (QUI)

- Prior to Practice Class 1: access and sign in at *QUIspace.org (using your unimelb email address)*
- In Practice Class: registered users -> expanded capabilities



MULT20015: subject summary 2024

24 Lectures:

Monday 11.00-12.00: PAR-115-L1-104-Medley Theatre (163)
Tuesday 14.15-15.15: PAR-115-L1-102-Latham Theatre (165)

12 Computer-based practice class tutorials (commencing in week 1):

Baker Computer Lab, Room 661, Level 6 David Caro Building (Physics)
Practice classes are scheduled on Thursday and Friday (see LMS)
QUI system – access at QUIspace.org, sign in with your unimelb email (for 10 qubit version)

Assessment (see Subject Outline on LMS for timing):

2 assignments (20%+30%), 2 hour exam (50%)

Staff (see LMS for contact details):

Subject coordinators: Dr. Josh Combes (Physics/Maths), Prof. Lloyd Hollenberg (Physics)

Lecturers: A/Prof. Muhammad Usman (Physics), Prof. Adrian Pearce (CIS), Dr. Nitin Yadav (Finance)

Head tutor: Chris Nakhl

LMS: Access lecture notes, practice class notes, assignments

Suggested reading:

E. Rieffel and W. Polak – "Quantum Computing: A Gentle Introduction"

IBM Quantum Experience – on-line textbook (qiskit.org/textbook)

Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8

Week 1

Lecture 1

1.1 A very brief overview of computing

1.2 The quantum world and quantum computing

Lecture 2

2.1 The mathematics of quantum states

2.2 Complex numbers and quantum amplitudes

2.3 Basic linear algebra: ket and matrix notation

2.4 State representation in the QUI

Tutorial 1

The Quantum User Interface (QUI), lecture review exercises

1.1 A very brief overview of computing

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Lecture 1

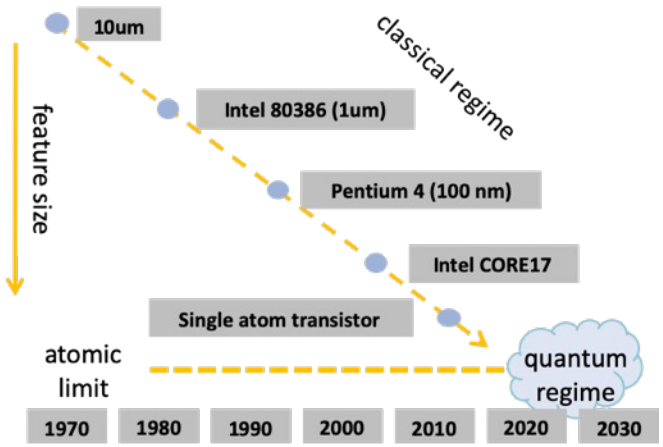
The limits of conventional computing

Conventional computers: some problems are just too hard to solve on even the biggest supercomputers -> More powerful CPUs?

Moore's Law – since 1960's transistor size halves every 1.5 years

But, as transistor size <10nm we're getting close to the limit (atoms are about 1nm or less)…

[also, they require increasing energy density to operate classically]



The end-of-Moore's law signals new computer technology is needed…

Many people believe quantum computers are the evolutionary step.

Quantum computers based on the laws of quantum mechanics circumvent limitations of classical information processing

The limits of conventional computing

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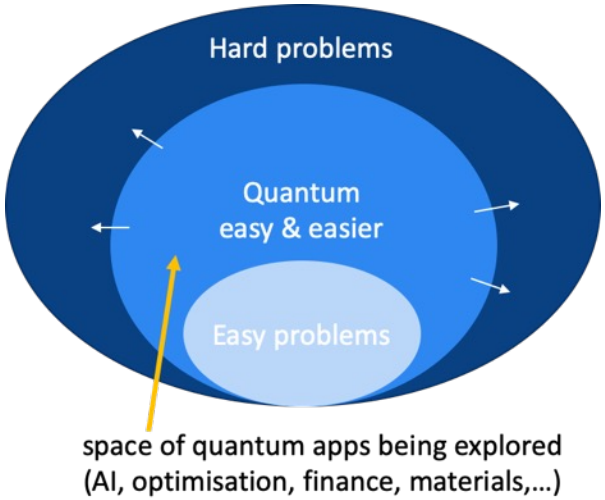

The end-of-Moore's law signals new computer technology is needed…

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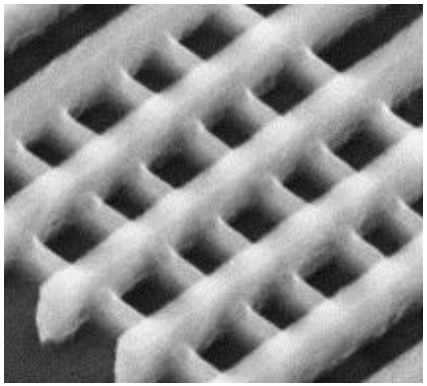
“Classical” computing: bits and binaries

Conventional computers obey the laws of “classical physics” (pre-quantum)

-> we call anything not quantum “classical”…i.e. “classical computers” vs. “quantum computers”

A classical bit (e.g. in an Intel chip) is like a switch with two values:



Intel.com

on “1”

off “0”

Only “0” <u>or</u> “1” at a time in any given physical representation (transistor)

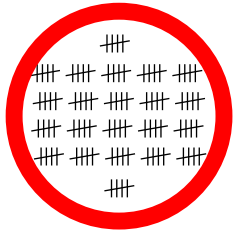
From classical bits → we can construct binary numbers 0000 <u>or</u> 1010 etc

Wait, what’s a binary number again?

Numbers…we need to re-learn some things

If you are like most people, you never stop to think about how numbers actually work…

Consider a humble speed sign:



c. 100,000 BC

Roman

Our number convention (left) is very efficient…

It is also decimal – based on powers of 10. What does this mean?

We know instinctively that "**110**" is shorthand for:

1x100 + **1**x10 + **0**x1 = "one hundred and one ten and no ones"

If we remember we're in base 10, we write it as:

$\mathbf{1} \times 10^2 + \mathbf{1} \times 10^1 + \mathbf{0} \times 10^0$

= one of 10^2 plus one of 10^1 plus zero of 10^0

$10^2 = 100$
$10^1 = 10$
$10^0 = 1$

"one hundred and ten"

"six"

But, in <u>base 2</u> the number "110" is actually "six", i.e.

Decimals and binaries

Let's break that down lest we get confused driving a car:

"one hundred and ten"

Decimal

$10^2 = 100$
$10^1 = 10$
$10^0 = 1$

$110 = 1 \times \underline{10}^2 + 1 \times \underline{10}^1 + 0 \times \underline{10}^0$ = one of $\underline{10}^2$ plus one of $\underline{10}^1$ plus zero of $\underline{10}^0$

= 100 + 10 + 0 = 110 as we normally read it

"six"

Binary

$2^2 = 4$
$2^1 = 2$
$2^0 = 1$

$110 = 1 \times \underline{2}^2 + 1 \times \underline{2}^1 + 0 \times \underline{2}^0$ = one of $\underline{2}^2$ plus one of $\underline{2}^1$ plus zero of $\underline{2}^0$

= 4 + 2 + 0 = 6

Hence: "There are 10 types of people in this world…

– those who understand binary numbers…and those who don't"

Some examples

Base 10 "decimal"

57

$57 = 5 \times 10^1 + 7 \times 10^0$ = five of 10^1 plus seven of $10^0 = 50 + 7 = 57$

Note: the numbers in blue run over <u>base 10</u> "digits" i.e. 0-9

57
10's 1's

$10^3 = 1000$
$10^2 = 100$
$10^1 = 10$
$10^0 = 1$

Roman

LVII

57 = L + VII = fifty plus two after five = 50 + 7 = 57

Note: incredibly cumbersome

100 = C
50 = L
4 = IV
1 = I

Base 2 "binary"

111001

$111001 = 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$= 1 \times 32 + 1 \times 16 + 1 \times 8 + 0 \times 4 + 0 \times 2 + 1 \times 1$

$= 32 + 16 + 8 + 0 + 0 + 1 = 57$

111001
32's 16's 8's 4's 2's 1's

$2^5 = 32$
$2^4 = 16$
$2^3 = 8$
$2^2 = 4$
$2^1 = 2$
$2^0 = 1$

Note: the numbers in blue run over <u>base 2</u> "bits" i.e. 0-1

The point: as computers naturally use binary numbers, so do quantum computers

→ need to be familiar with bits and binary and decimals (see table)

Decimal	Binary	Decimal	Binary
0	000	4	100
1	001	5	101
2	010	6	110
3	011	7	111

“Classical” computing: bits and binaries

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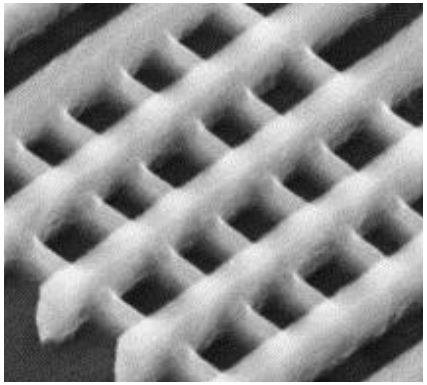

Intel.com

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Only “0” <u>or</u> “1” at a time in any given physical representation (transistor)

Continuing:

From classical bits → we can construct binary numbers 0000 <u>or</u> 1010 etc

One binary number at one time in the physical representation

Quantum information is very different…

1.2 The quantum world and quantum computing

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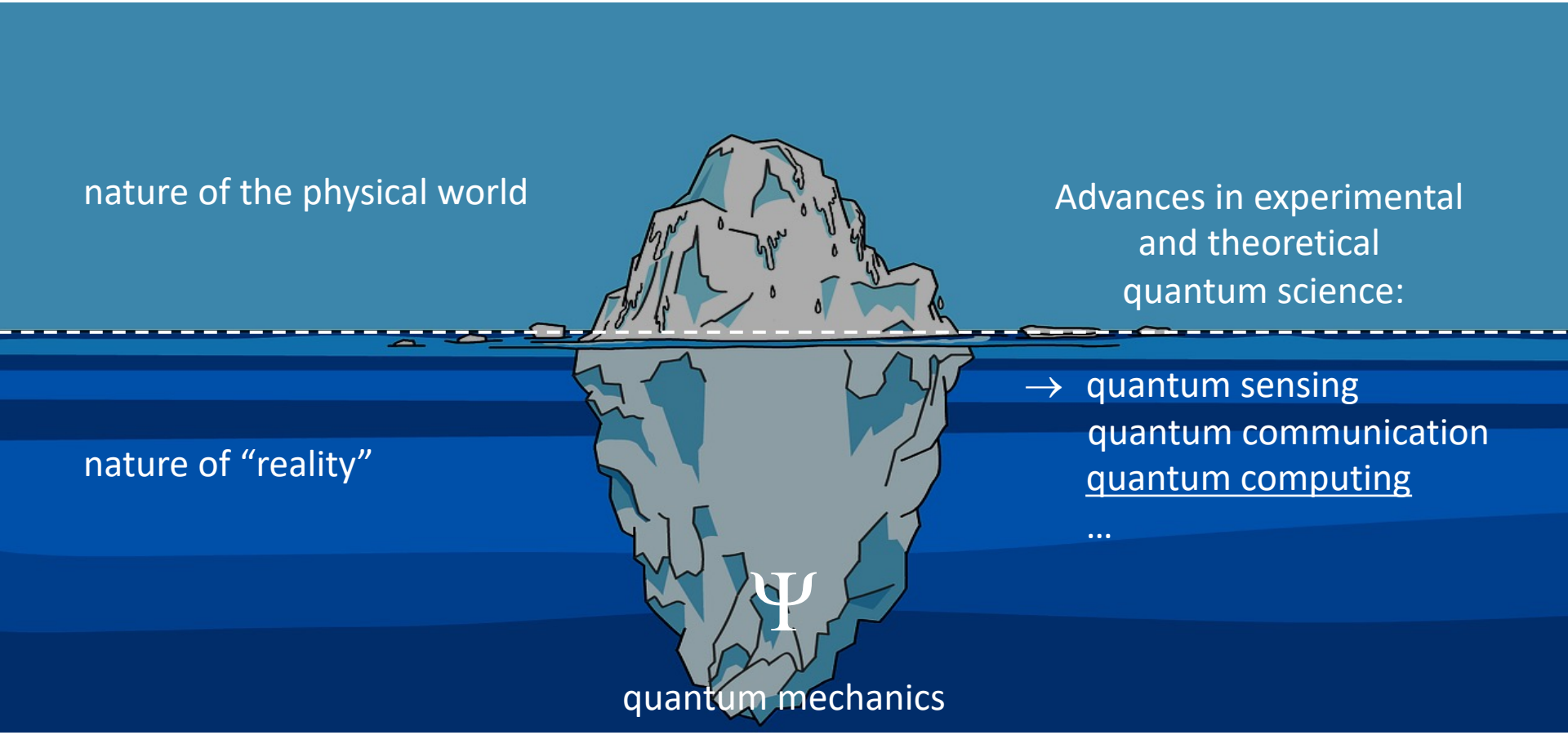
Lecture 1

The Quantum World

Some meta-physics...



Key concepts for quantum computing

Quantum superposition:

Systems can be in indeterminate (multiple) states prior to measurement

Quantum measurement:

Result of any given measurement a-priori unknown, system “collapses” to an outcome

Quantum entanglement:

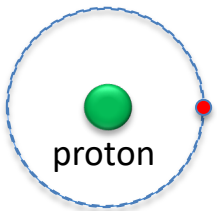
Systems can be linked such that measurement of one part correlates to that of another part

We’ll go through each of these systematically…

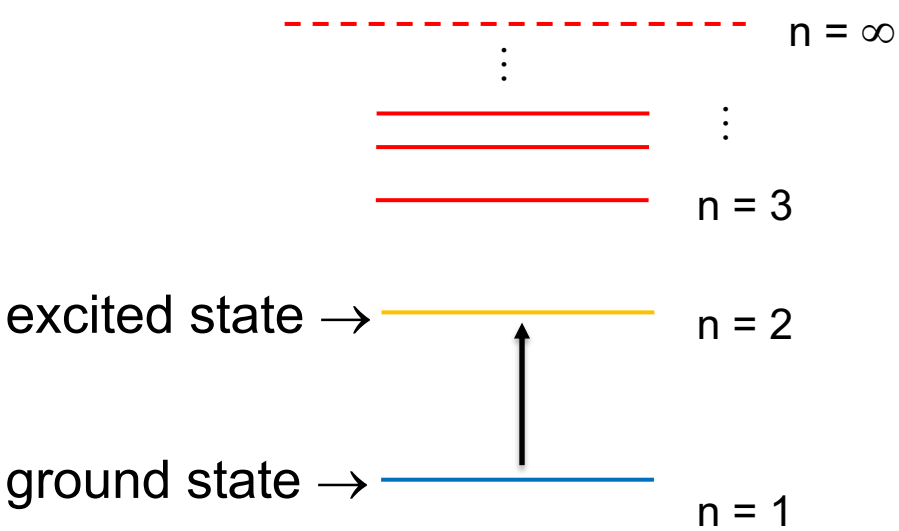
Start with a (semi) familiar concept – the atom

Planck (1900): postulated that energy is quantised

Bohr (1913): constructed a simple "quantised" model of the hydrogen atom



$$E_n = -\frac{m}{2\hbar^2}\left(\frac{e^2}{4\pi\,\varepsilon_0}\right)^2 \frac{1}{n^2}$$



energy levels

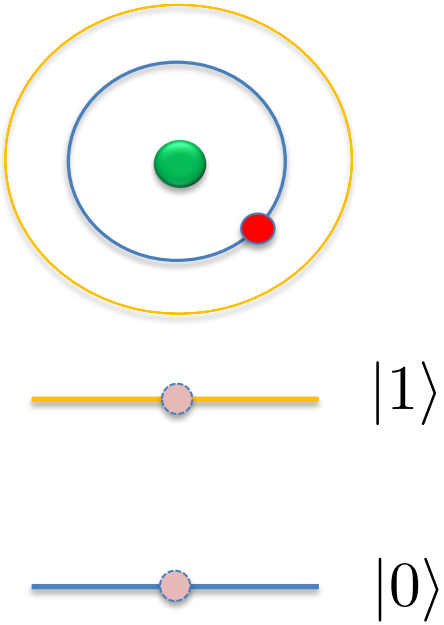
Consider lowest two levels

Simply re-label states to a bit notation

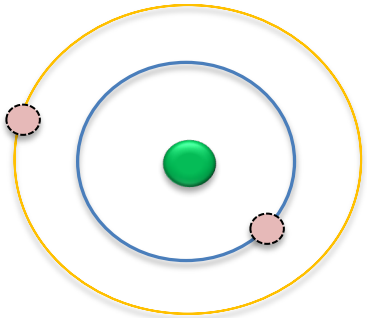
→ quantum bit, or "qubit"

(Not to be confused with "cubit"!)

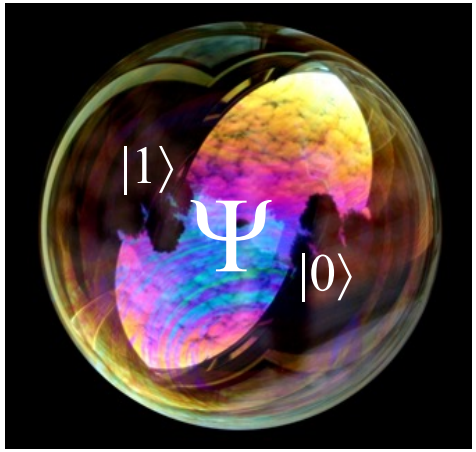
Quantum superposition and measurement



One electron in quantum superposition.

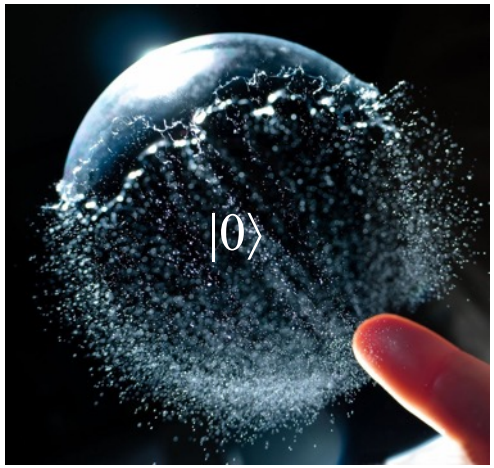


quantum superposition



$|0\rangle$ “and” $|1\rangle$

measurement/observation



state collapse $\rightarrow$ random outcome

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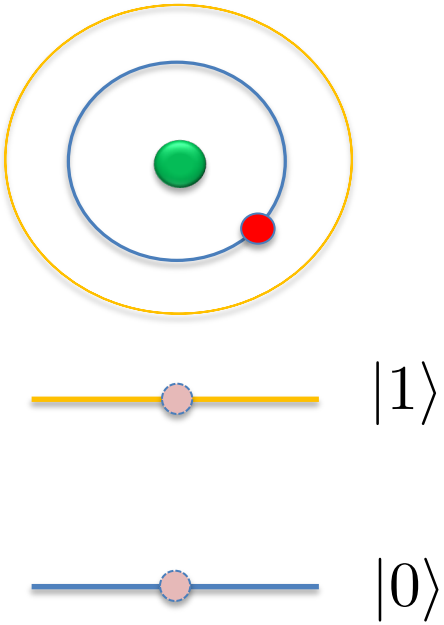
Quantum superposition and measurement



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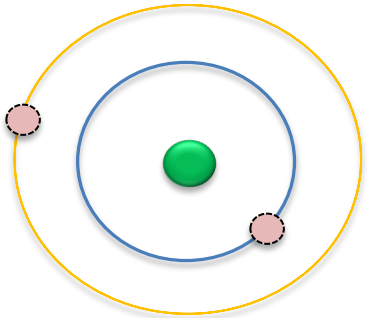
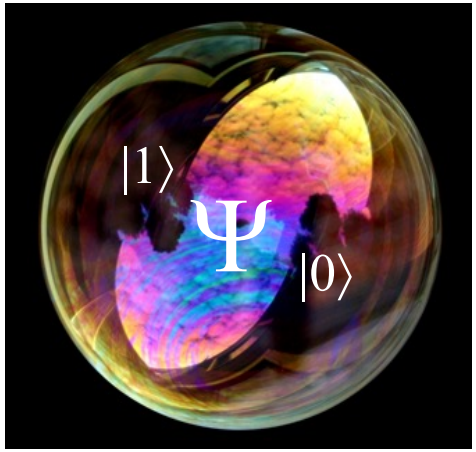


quantum superposition



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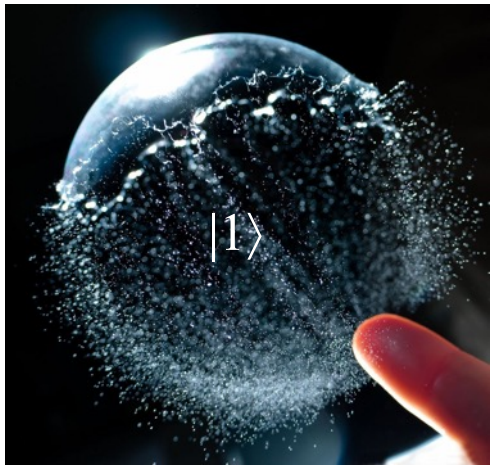

state collapse $\rightarrow$ random outcome

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Quantum entanglement

Imagine two qubits. We can prepare two distinct classes of overall state:

a) Separable state: independent quantum superpositions

qubit-1 $\sim |0\rangle + |1\rangle$
qubit-2 $\sim |0\rangle + |1\rangle$

(NB for later: ignoring normalisation for now)

Or we write $(|0\rangle + |1\rangle)$ x $(|0\rangle + |1\rangle) \rightarrow |0\rangle\,|0\rangle + |0\rangle\,|1\rangle + |1\rangle\,|0\rangle + |1\rangle\,|1\rangle \rightarrow |00\rangle + |01\rangle + |10\rangle + |11\rangle$

Measurement outcomes for each qubit: random and independent

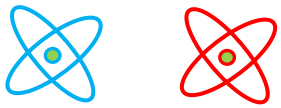
b) An entangled state: e.g. in above notation $|0\rangle\,|1\rangle + |1\rangle\,|0\rangle$

Measurement on qubit-1:
qubit-1 = $|0\rangle$ (random) $\rightarrow$ qubit-2 = $|1\rangle$
qubit-1 = $|1\rangle$ (random) $\rightarrow$ qubit-2 = $|0\rangle$

(and vice-versa if we first measured qubit-2)

Measurement outcomes for e.g. qubit-1 are random, but result of qubit-2 depends on qubit-1 outcome $\rightarrow$ the qubits are somehow connected…

Multi qubits – state counting

		Binary combinations
1 qubit:	$\lvert 0\rangle, \lvert 1\rangle$	$\lvert 0\rangle, \lvert 1\rangle$ binary representation of decimals 0 to 1
2 qubits:	 $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$	$\lvert 00\rangle, \lvert 01\rangle, \lvert 10\rangle, \lvert 11\rangle$ binary representation of decimals 0 to 3
3 qubits:	$\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$	$\lvert 000\rangle, \lvert 001\rangle, \lvert 010\rangle, \lvert 011\rangle$ $\lvert 100\rangle, \lvert 101\rangle, \lvert 110\rangle, \lvert 111\rangle$ binary representation of decimals 0 to 7
4 qubits:	$\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$	$\lvert 0000\rangle, \lvert 0001\rangle, \lvert 0010\rangle, \lvert 0011\rangle$ $\lvert 0100\rangle, \lvert 0101\rangle, \lvert 0110\rangle, \lvert 0111\rangle$ $\lvert 1000\rangle, \lvert 1001\rangle, \lvert 1010\rangle, \lvert 1011\rangle$ $\lvert 1100\rangle, \lvert 1101\rangle, \lvert 1110\rangle, \lvert 1111\rangle$ binary representation of decimals 0 to 15

Multiple qubits and quantum processing

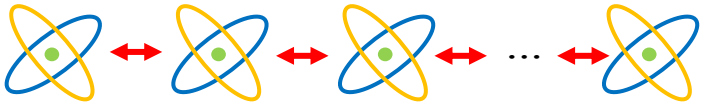
Basic <u>representation</u> of binaries as quantum information:

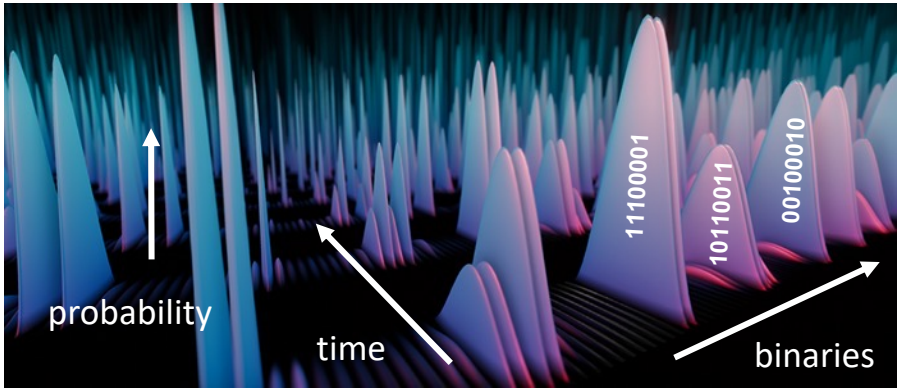
N qubits

Independent quantum superpositions → superposition over N-bit binaries $|000\ldots0\rangle, \ldots, |111\ldots1\rangle$ (and there are 2^N of these)

Not very useful…measurement of qubits collapses to one random N-bit string

<u>Quantum computation</u>: qubits interact to create complex superpositions and entangled states





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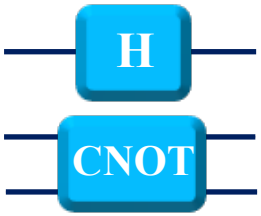
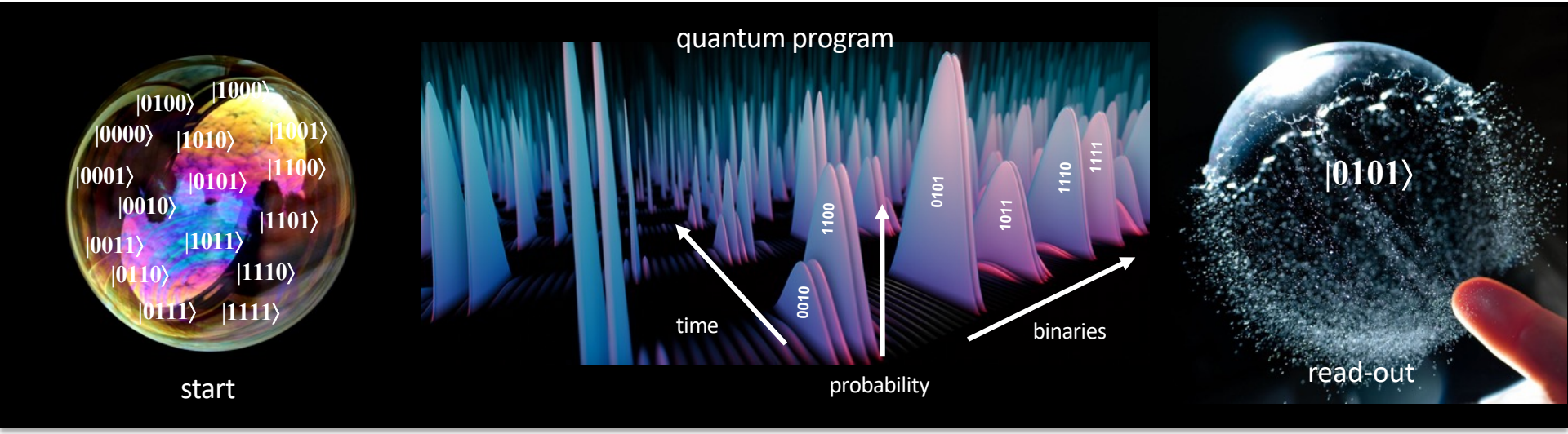
Quantum information processing

Digital quantum computing:

- logic gates between qubits perform mathematical operations on binary data
- complex entangled states created → binary data are quantum "linked"
- quantum interference amplifies probability of desired output (answer)





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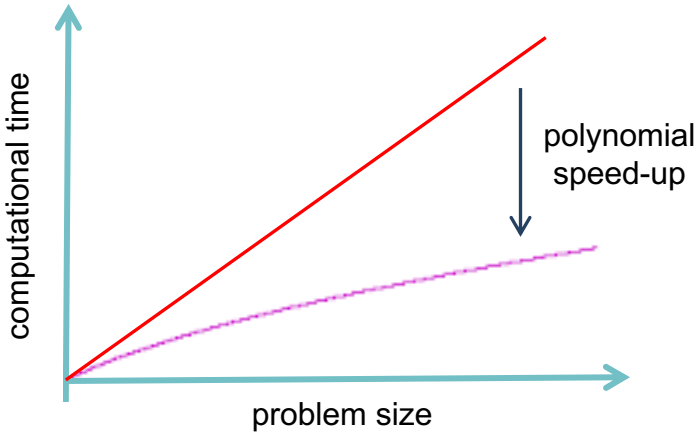
Types of quantum computers

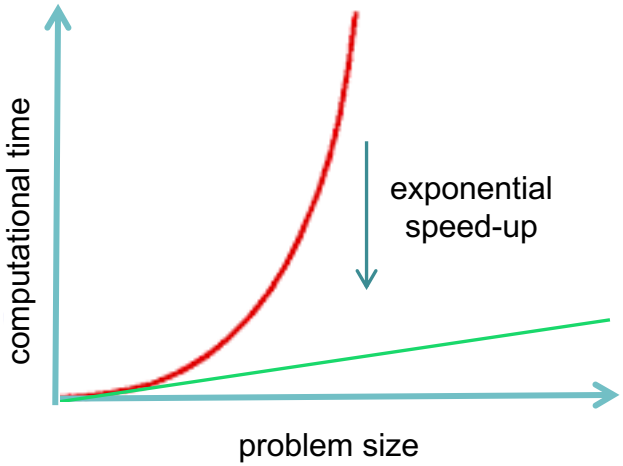
Broadly, there are two classes of *digital* quantum computers*:

- **Intermediate quantum computers (IQC)**
  - medium-scale system (10 ~1000 qubits)
  - simplified control and error correction
  - potentially polynomial speed-up (e.g. √CPU)
  - optimisation, machine learning, chemistry,…
  - pathway to full-scale universal QC…

  (NISQ = "Noisy Intermediate Scale Quantum")



- **Full universal quantum computers (UQC)**
  - large-scale system (>>1000 qubits)
  - high redundancy for quantum error correction
  - potentially exponential speed-up for some problems (e.g. Shor's factoring algorithm)
  - large class of problems: financial, data-base analysis, security, bio-molecular simulation
  - polynomial to exponential speed-up

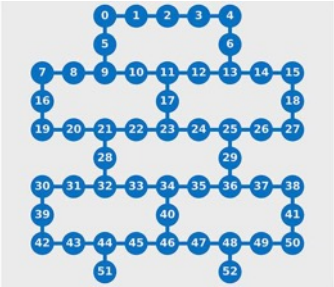


*c/f "analogue" machines for Adiabatic Quantum Computing, such as D-Wave systems

Meanwhile: quantum computers emerge

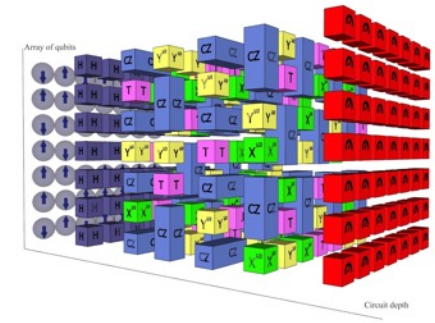
2016: IBM provides cloud access to QC hardware, programming interface
2017: IBM Network Q (UoM a founding Hub)
2019: Quantum supremacy
2023: Quantum utility?

Other players: Rigetti, Google, IonQ, Microsoft, Intel, D-Wave,…



2019: "System One" IBM state-of-the-art
Stand-alone QC systems (20 → 65 qubits),
fully programmable [2023 -> 400-qubit level]

IBM Quantum Network Hub
at the University of Melbourne



Google.com: quantum circuit sampler machine

2019: Google achieves "Quantum Supremacy"
53 qubits vs. HPC for random circuits 200 sec (QC)
vs. 10,000 yrs (HPC) [Nature Oct 23 2019]

2020: Hefei photonic QC
100-qubit equiv. vs. HPC for Boson Sampling
200 sec (QC) vs. 600 m y [Science Dec 03 2020]

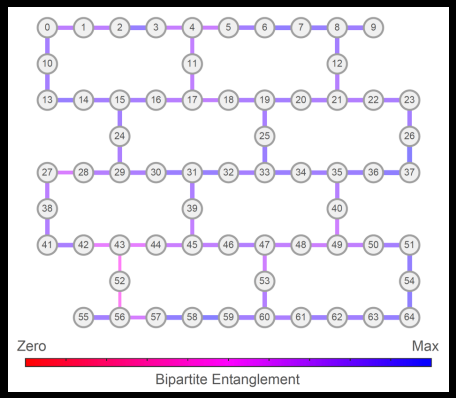
2023: IBM Quantum Utility
Sim. 2D transverse-field Ising model on 127 qubits
[Nature Jun 14 (2023)]

Big goal: "Quantum advantage"

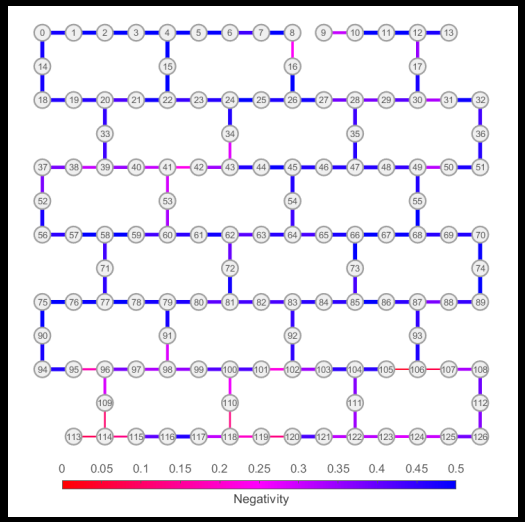
i.e. beat HPC on a <u>useful</u> problem (if/when?)

Aside: UoM whole-QC entanglement (65 -> 127 qubits)

Generation and verification of the largest entangled “Graph States” -> potential for benchmarking whole-device



$$|G_n\rangle = \prod_{(a,b)\in E} \mathrm{CZ}_b^a |+\rangle^{\otimes n}$$



Mooney et al (2021): 65-qubit entangled state (whole device)

G. Mooney, G. White, C. Hill, LH, Adv. Q Tech. 4 2100061 (2021)

Kam et al (MSc, 2022): whole-device entanglement on a 127 qubit system (*ibmq_washington*)

Aside: Introducing IBM's 414 qubit quantum computer…

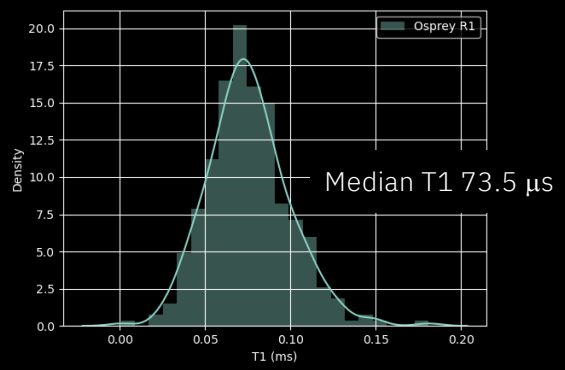
IBM Osprey - First Results

2019
Falcon
27 Qubits

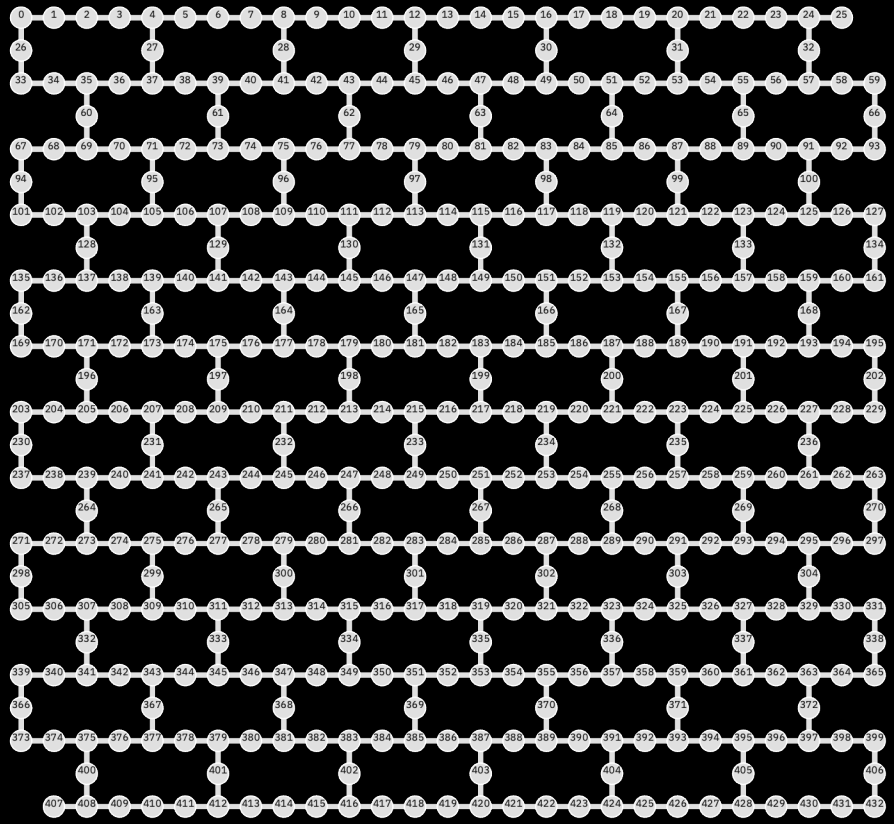
2020
Hummingbird
65 Qubits

2021
Eagle
127 Qubits

2022
Osprey
433 Qubits



Osprey connectivity map

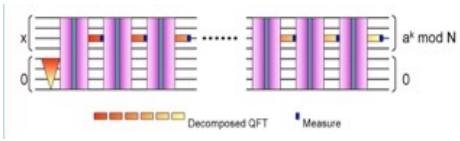


D. Gill/J. Gambetta et al, IBM.com

Timeline and state of the art

Google: “Quantum Supremacy”, Hefei: Boson Sampling problem

Quantum advantage 100-1000 qubits?



increasing # qubits and quality (error rates)

Universal QC: (millions of qubits) poly to exponential speed-up on various problems

Theory: large-scale simulations c.50 qubits

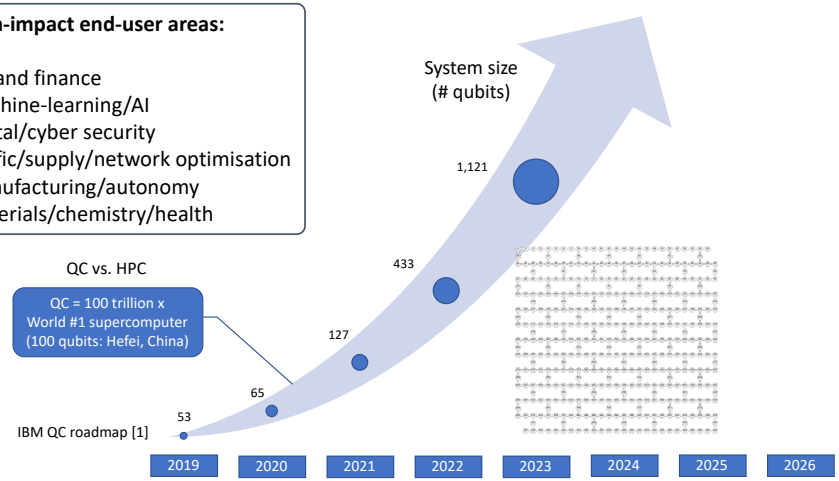
Hardware race: IBM, Google, Intel, D-Wave, Rigetti, Microsoft, SQC,…

DWave: 2000 “qubits” -> <u>analogue</u> QC

IBM: 400+ qubits -> <u>digital</u> QC

High-impact end-user areas:

ICT and finance
Machine-learning/AI
Digital/cyber security
Traffic/supply/network optimisation
Manufacturing/autonomy
Materials/chemistry/health



In the meantime, the era of quantum software and app development has begun…

Week 1

Lecture 1

1.1 A very brief overview of computing

1.2 The quantum world and quantum computing

Lecture 2

2.1 The mathematics of quantum states

2.2 Complex numbers and quantum amplitudes

2.3 Basic linear algebra: ket and matrix notation

2.4 State representation in the QUI

Tutorial 1

The Quantum User Interface (QUI), lecture review exercises

Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8